

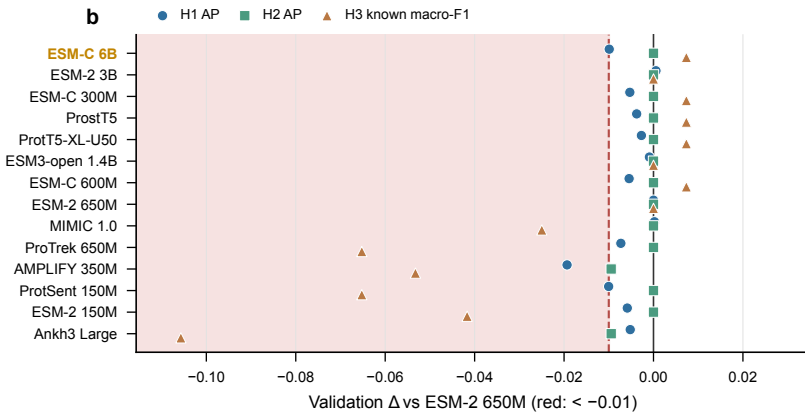
Development-only selection of protein representations

Frozen Train-only 5-fold global-component CV (mean $\pm$ fold SE)

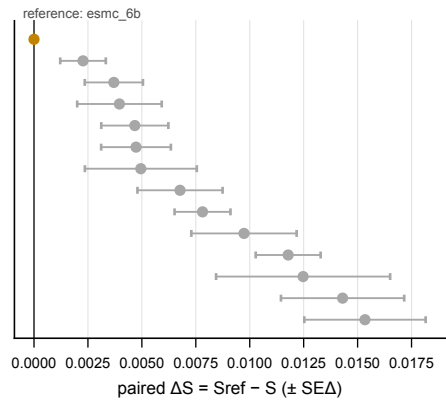
data-curation V3 (560 viral DJR-MCPs) $\rightarrow$ project V0 $\cdot$ raw-score-stable-calibration-v1 $\cdot$ Train CV + Validation only $\cdot$ no Test metrics

a	S	H1 AP	H2 AP	H3 known macro-F1
ESM-C 6B	0.997 ± 0.001	0.998 ± 0.000	1.000 ± 0.000	0.981 ± 0.010
ESM-2 3B	0.995 ± 0.001	0.999 ± 0.000	1.000 ± 0.000	0.953 ± 0.007
ESM-C 300M	0.993 ± 0.002	0.995 ± 0.001	1.000 ± 0.000	0.962 ± 0.017
ProstT5	0.993 ± 0.002	0.996 ± 0.003	1.000 ± 0.000	0.957 ± 0.009
ProtT5-XL-U50	0.992 ± 0.002	0.997 ± 0.001	1.000 ± 0.000	0.944 ± 0.013
ESM3-open 1.4B	0.992 ± 0.001	0.995 ± 0.002	1.000 ± 0.000	0.952 ± 0.008
ESM-C 600M	0.992 ± 0.003	0.995 ± 0.003	1.000 ± 0.000	0.950 ± 0.016
ESM-2 650M	0.990 ± 0.002	0.998 ± 0.001	1.000 ± 0.000	0.916 ± 0.025
MIMIC 1.0	0.989 ± 0.002	0.984 ± 0.001	1.000 ± 0.000	0.930 ± 0.016
ProTrek 650M	0.987 ± 0.003	0.995 ± 0.002	1.000 ± 0.000	0.907 ± 0.028
AMPLIFY 350M	0.985 ± 0.002	0.988 ± 0.001	0.999 ± 0.000	0.886 ± 0.022
ProtSent 150M	0.985 ± 0.004	0.988 ± 0.004	1.000 ± 0.000	0.919 ± 0.024
ESM-2 150M	0.982 ± 0.003	0.991 ± 0.004	1.000 ± 0.000	0.876 ± 0.018
Ankh3 Large	0.982 ± 0.002	0.991 ± 0.002	1.000 ± 0.000	0.872 ± 0.014

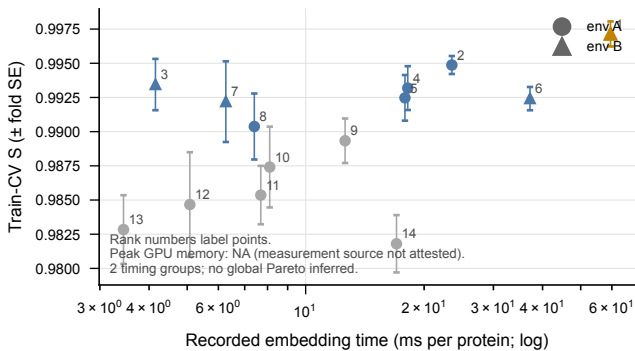
Validation gates



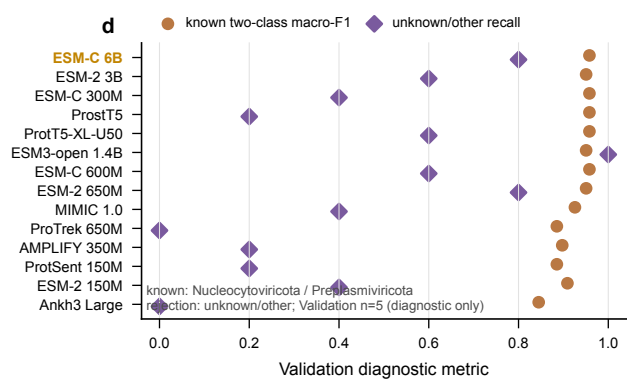
Paired one-SE



S—compute evidence (descriptive)



H3 output and rejection scope



Frozen decision audit — selected ESM-C 6B (baseline = ESM-2 650M)

Model	1-SE	mean $\Delta S \pm SE_{\Delta}$	Val gate	License	Pretraining-overlap risk	Decision
ESM-C 6B	yes	$+0.000 \pm 0.000$	PASS	MIT	not flagged	selected; tie-break rank 1
ESM-2 3B	no	$+0.002 \pm 0.001$	PASS	MIT	not flagged	outside paired one-SE set
ESM-C 300M	no	$+0.004 \pm 0.001$	PASS	MIT	not flagged	outside paired one-SE set
ProstT5	no	$+0.004 \pm 0.002$	PASS	MIT	high	outside paired one-SE set
ProtT5-XL-U50	no	$+0.005 \pm 0.002$	PASS	Apache-2.0	not flagged	outside paired one-SE set
ESM3-open 1.4B	no	$+0.005 \pm 0.002$	PASS	MIT	low (viruses excluded)	outside paired one-SE set
ESM-C 600M	no	$+0.005 \pm 0.003$	PASS	MIT	not flagged	outside paired one-SE set
ESM-2 650M	no	$+0.007 \pm 0.002$	PASS	MIT	not flagged	outside paired one-SE set
MIMIC 1.0	no	$+0.008 \pm 0.001$	FAIL H3	MIT	high	excluded by Validation H3
ProTrek 650M	no	$+0.010 \pm 0.002$	FAIL H3	MIT	high	excluded by Validation H3
AMPLIFY 350M	no	$+0.012 \pm 0.002$	FAIL H1, H3	MIT	not flagged	excluded by Validation H1, H3
ProtSent 150M	no	$+0.012 \pm 0.004$	FAIL H1, H3	MIT	high	excluded by Validation H1, H3
ESM-2 150M	no	$+0.014 \pm 0.003$	FAIL H3	MIT	not flagged	excluded by Validation H3
Ankh3 Large	no	$+0.015 \pm 0.003$	FAIL H3	CC-BY-NC-SA-4.0	not flagged	excluded by Validation H3